Business Case Study - Aerofit

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
from google.colab import files
uploaded = files.upload()
```

Choose Files Aerofit.csv
Aerofit.csv(text/csv) - 7460 bytes, last modified: 2/23/2026 - 100% done
```
Saving Aerofit.csv to Aerofit.csv
```

Imported the data into colab book

```
df = pd.read_csv('/content/Aerofit.csv')
df.head()
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
0	KP281	18	Male	14	Single	3	4	29562	112
1	KP281	19	Male	15	Single	2	3	31836	75
2	KP281	19	Female	14	Partnered	4	3	30699	66
3	KP281	19	Male	12	Single	3	3	32973	85
4	KP281	20	Male	13	Partnered	4	2	35247	47

Next steps: Generate code with df New interactive sheet

Aerofit dataset information

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180 entries, 0 to 179
Data columns (total 9 columns):
 #   Column         Non-Null Count  Dtype 
---  ------         --------------  ----- 
 0   Product        180 non-null    object
 1   Age            180 non-null    int64 
 2   Gender         180 non-null    object
 3   Education      180 non-null    int64 
 4   MaritalStatus  180 non-null    object
 5   Usage          180 non-null    int64 
 6   Fitness        180 non-null    int64 
 7   Income         180 non-null    int64 
 8   Miles          180 non-null    int64 
dtypes: int64(6), object(3)
memory usage: 12.8+ KB
```

Aerofit data consist of 180 product purchase data of differnt customers data set has 180 rows and 11 columns

Added below Colmns for analysis Purpose

1. Product_tier
2. price
3. Age_group bin

```
df['Product_Tier'] = df['Product'].map({
    'KP281': 'Entry',
    'KP481': 'Mid',
    'KP781': 'Premium'
})
```

```
df['Price'] = df['Product'].map({
    'KP281': 1500,
    'KP481': 1750,
    'KP781': 2500
})
```

```
df['Age_group'] = pd.cut(df['Age'], bins=[17,25,35,40,60] , labels=['17-25','26-35','36-40','41-60'])
```

```
df['Usage_Level'] = pd.cut(df['Usage'],
                           bins=[0,2,4,7],
                           labels=['Low','Medium','High'])
```

```
df['Fitness_Level'] = pd.cut(
    df['Fitness'],
    bins=[0,2,3,5],
```

```
    labels=['Low','Medium','High'],
    include_lowest=True
)
```

```
df.head()
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles	Product_Tier	Price	Age_group	Usage_Level	Fitness_Level
0	KP281	18	Male	14	Single	3	4	29562	112	Entry	1500	17-25	Medium	High
1	KP281	19	Male	15	Single	2	3	31836	75	Entry	1500	17-25	Low	Medium
2	KP281	19	Female	14	Partnered	4	3	30699	66	Entry	1500	17-25	Medium	Medium
3	KP281	19	Male	12	Single	3	3	32973	85	Entry	1500	17-25	Medium	Medium
4	KP281	20	Male	13	Partnered	4	2	35247	47	Entry	1500	17-25	Medium	Low

Next steps: Generate code with df New interactive sheet

```
df['Product'].value_counts()
```

Product	count
KP281	80
KP481	60
KP781	40

dtype: int64

```
df['Age_group'].value_counts()
```

Age_group	count
17-25	79
26-35	73
36-40	16
41-60	12

dtype: int64

```
df['Product'].unique()
```

```
array(['KP281', 'KP481', 'KP781'], dtype=object)
```

```
df['Product_Tier'].unique()
```

```
array(['Entry', 'Mid', 'Premium'], dtype=object)
```

```
df['Gender'].unique()
```

```
array(['Male', 'Female'], dtype=object)
```

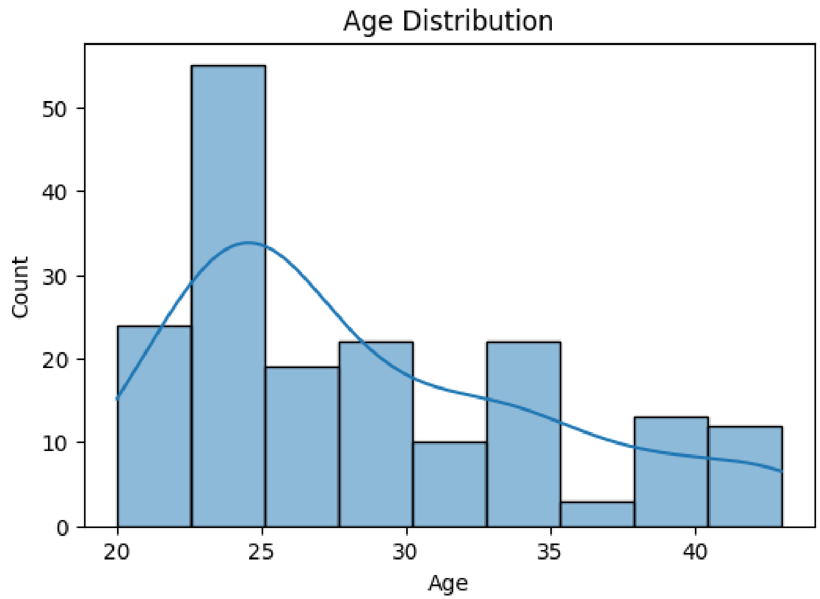
Cleaning data

Decimal values in Age , Useage and Income rounding the value to Complete int

```
df['Income'] = df['Income'].round().astype(int)
```

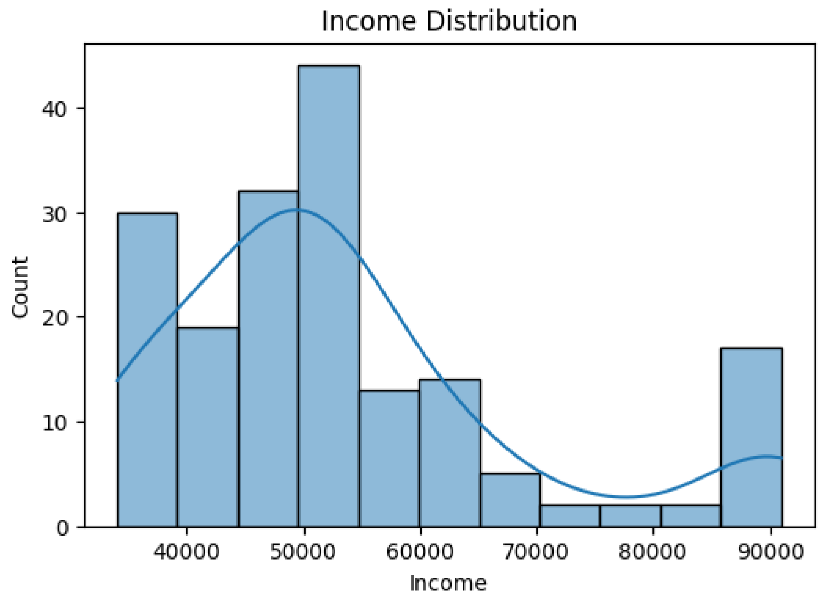
Fig Analysis : Univeriate and Biveriate analysis

```
#Age group distibution

plt.figure(figsize=(6,4))
sns.histplot(df['Age'], kde=True)
plt.title("Age Distribution")
plt.show()
```

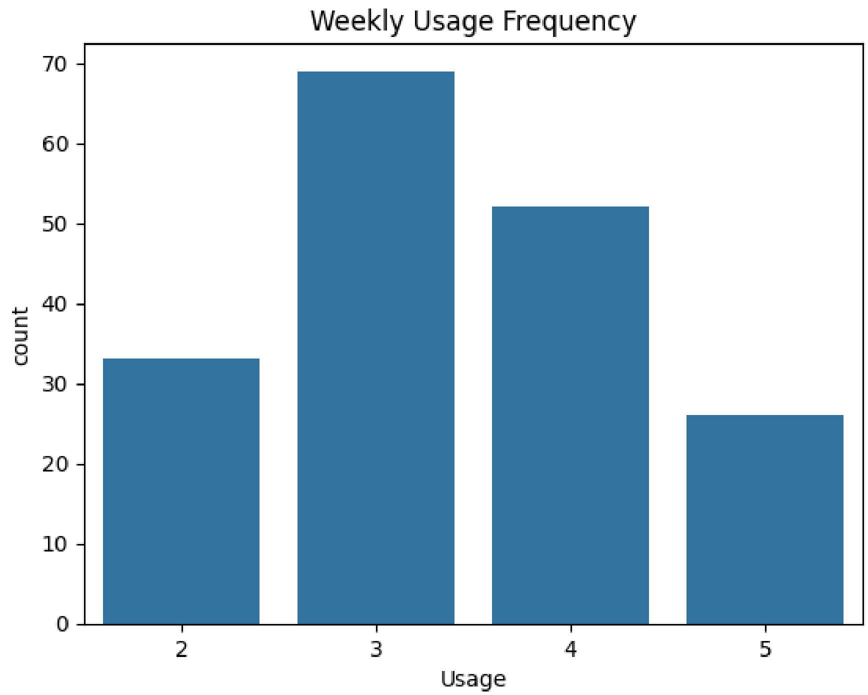


Double-click (or enter) to edit

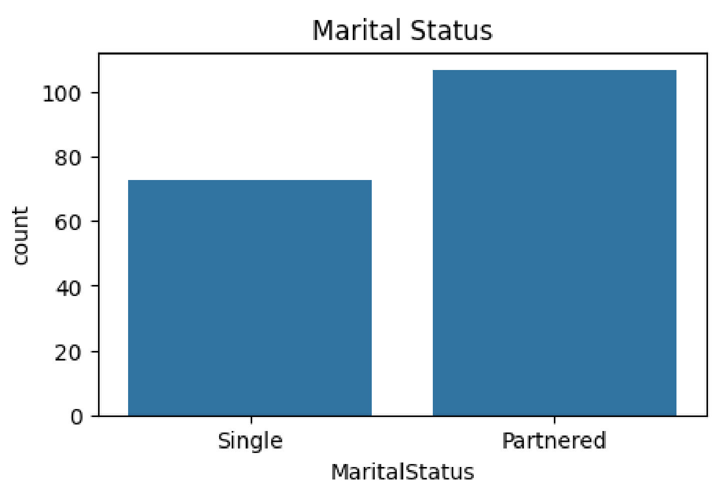
```
#Income distribution

plt.figure(figsize=(6,4))
sns.histplot(df['Income'], kde=True)
plt.title("Income Distribution")
plt.show()
```




```
sns.countplot(x='Usage', data=df)
plt.title("Weekly Usage Frequency")
plt.show()
```

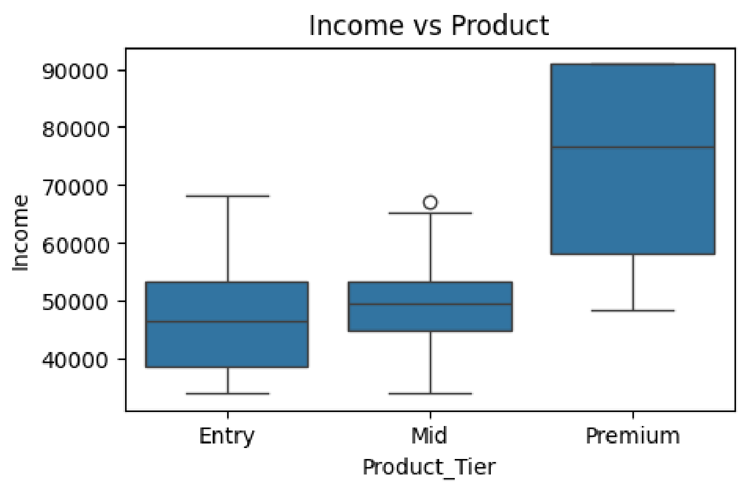



```
plt.figure(figsize=(5,3))
sns.countplot(x='MaritalStatus', data=df)
plt.title("Marital Status")
plt.show()
```



Below figer shows that those who have higher income buy premium product range

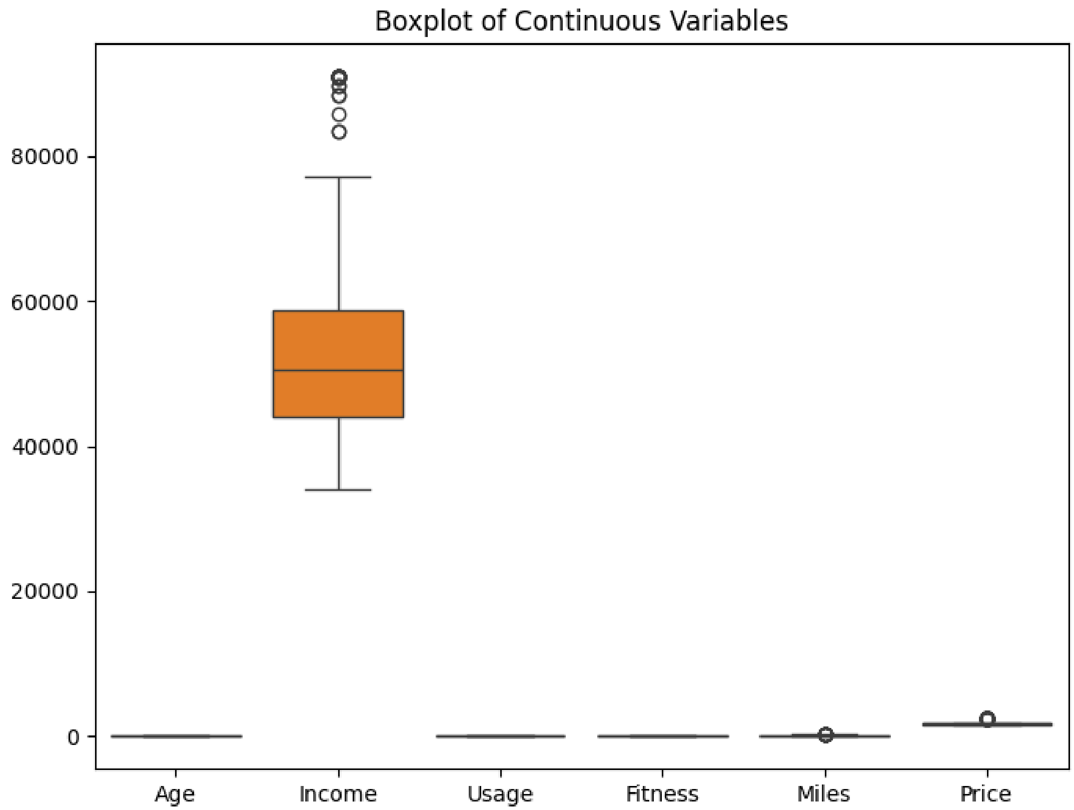
```
plt.figure(figsize=(5,3))
sns.boxplot(x='Product_Tier', y='Income', data=df)
plt.title("Income vs Product")
plt.show()
```



Defining Outliers amoung columns

```
continuous_cols = ['Age','Income','Usage','Fitness','Miles','Price']
```

```
plt.figure(figsize=(8,6))
sns.boxplot(data=df[continuous_cols])
plt.title("Boxplot of Continuous Variables")
plt.show()
```




```
for col in continuous_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    outliers = df[(df[col] < lower) | (df[col] > upper)]

    print(f"{col}: {len(outliers)} outliers")
```

```
Age: 0 outliers
Income: 19 outliers
Usage: 0 outliers
Fitness: 0 outliers
Miles: 13 outliers
Price: 40 outliers
```

Removing Outliers below 5% and above 95%

```
lower = df['Income'].quantile(0.05)
upper = df['Income'].quantile(0.95)
```

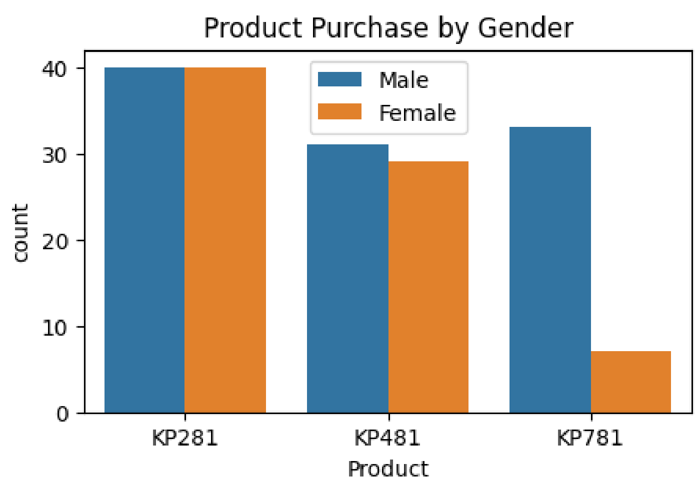
```
for col in continuous_cols:
    lower = df[col].quantile(0.05)
    upper = df[col].quantile(0.95)
    df[col] = np.clip(df[col], lower, upper)
```

```
df['Price'].max()
```

```
2500
```

Finding Relation between Categorical column and output

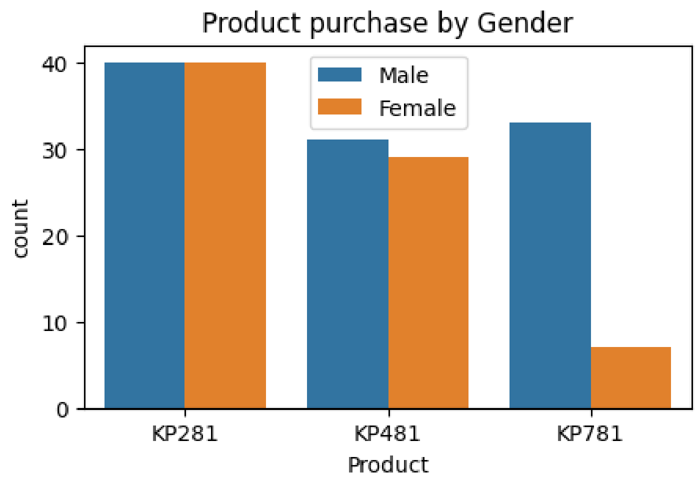
```
#Gender vs Product
plt.figure(figsize=(5,3))
sns.countplot(x='Product', hue='Gender', data=df )
plt.title("Product Purchase by Gender")
plt.legend()
plt.show()
```



```
df.columns
```

```
Index(['Product', 'Age', 'Gender', 'Education', 'MaritalStatus', 'Usage',
       'Fitness', 'Income', 'Miles', 'Product_Tier', 'Price', 'Age_group',
       'Usage_Level', 'Fitness_Level'],
      dtype='object')
```

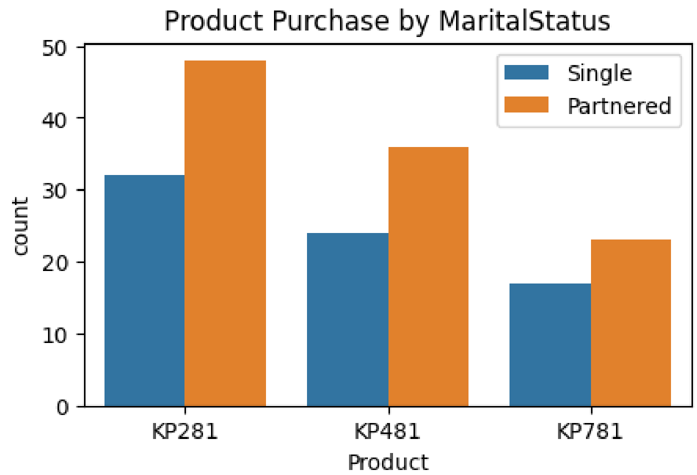
```
plt.figure(figsize=(5,3))
sns.countplot(x= 'Product' , hue = 'Gender' , data = df)
plt.title("Product purchase by Gender")
plt.legend()
plt.show()
```




```
pd.crosstab(df['Gender'], df['Product'], normalize='index') * 100
```

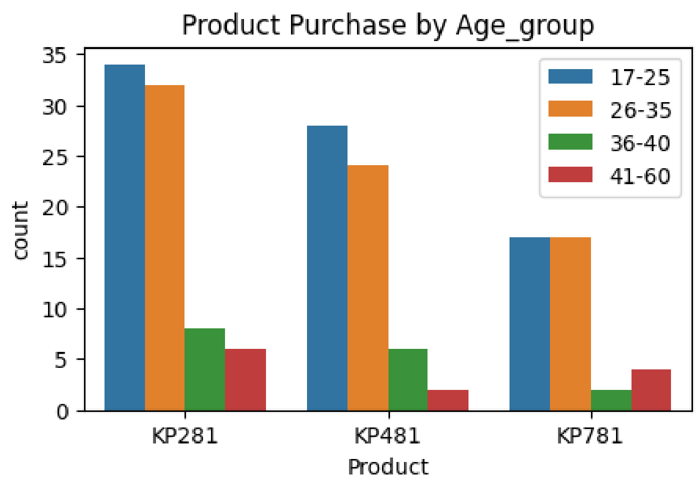
Product Gender	KP281	KP481	KP781
Female	52.631579	38.157895	9.210526
Male	38.461538	29.807692	31.730769

```
#Gender vs Product
plt.figure(figsize=(5,3))
sns.countplot(x='Product', hue='MaritalStatus',  data = df )
plt.title("Product Purchase by MaritalStatus")
plt.legend()
plt.show()
```



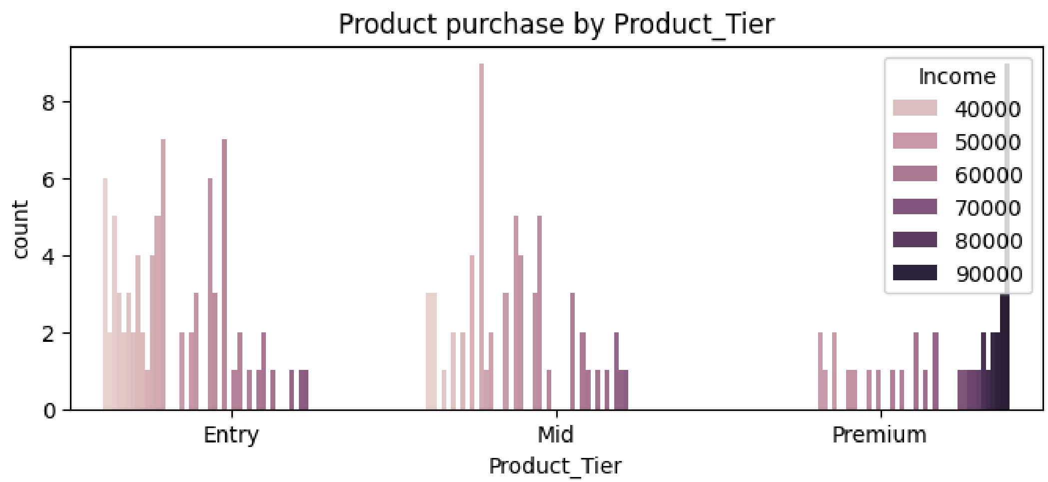
Below figure shows which age group buy which product

```
plt.figure(figsize=(5,3))
sns.countplot(x='Product' , hue = 'Age_group' , data = df)
plt.title("Product Purchase by Age_group")
plt.legend()
plt.show()
```

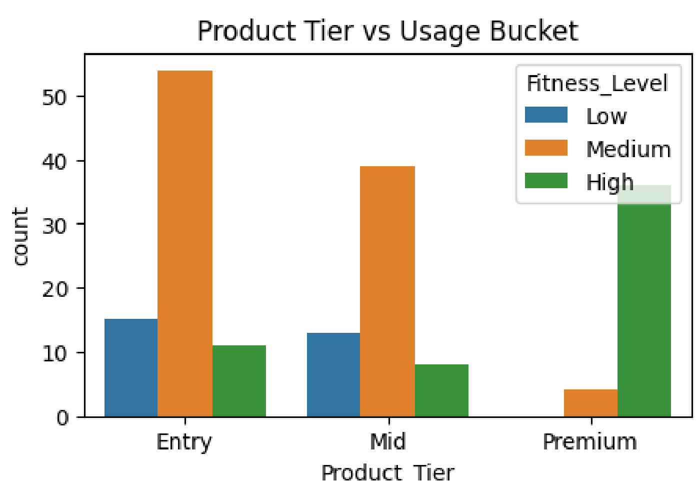



```
plt.figure(figsize=(8,3))
sns.countplot(x ='Product_Tier' , hue= 'Income' , data = df)
```

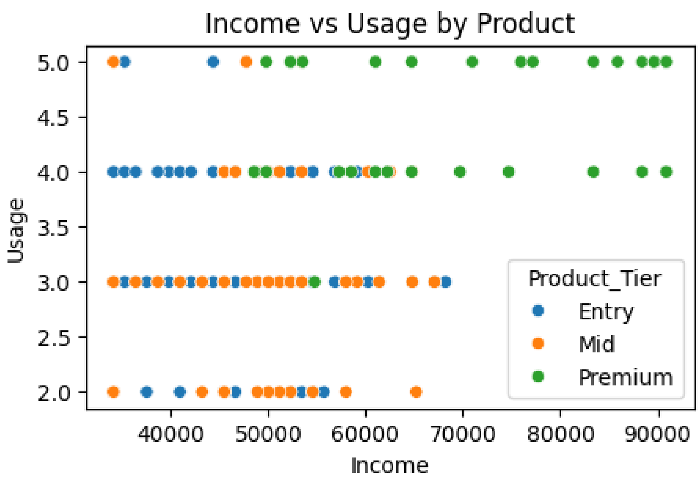
```
plt.title("Product purchase by Product_Tier")
plt.show()
```




```
plt.figure(figsize=(5,3))
sns.countplot(
    data=df,
    x='Product_Tier',
    hue='Fitness_Level'
)
plt.title("Product Tier vs Usage Bucket")
plt.show()
```



```
plt.figure(figsize=(5,3))
sns.scatterplot(x='Income', y='Usage',
                hue='Product_Tier',
                data=df)
plt.title("Income vs Usage by Product")
plt.show()
```



From the above analysis we observed that

1. Male Customers prefer buy Premium product
2. Younger age group prefer entry level product
3. Partnered section slightly buy mid to premium level product
4. Customers with higher fitness level and weekly usage by premium products

```
df.to_csv('Aerofit.csv', index=False)

from google.colab import files
files.download('Aerofit.csv')
```

Representation of probability

```
#Overall marginal Probability for customer to purchase product

marginal = (pd.crosstab(df['Product'] , columns= 'Probability' , normalize= True)*100).reset_index()
```

```
marginal.columns = ['Product' , 'Probability' ]

marginal
```

	Product	Probability
0	KP281	44.444444
1	KP481	33.333333
2	KP781	22.222222

Next steps: Generate code with marginal New interactive sheet

```
marginal_G = pd.crosstab(df['Gender'],df['Product'] ,normalize='index') * 100

marginal_G.columns.name = None

marginal_G.index.name = None

marginal_G = marginal_G.reset_index()

marginal_G
```

	index	KP281	KP481	KP781
0	Female	52.631579	38.157895	9.210526
1	Male	38.461538	29.807692	31.730769

Next steps: Generate code with marginal_G New interactive sheet

Findings

1. Overall marginal probability shows the KP281 has the higthes probability to be purchased amoung all customers
2. Male customers has high probability to purchase KP281 (39%) which is mostly by low to mid income male customers followed by KP781 (31%) which is populer amount high income male customers
3. Female customers has high probability to purchase KP281 (52%) Followed by KP481 (38%)

Correlation

```
numeric_cols = ['Age','Education','Usage','Fitness','Income','Miles','Price'  ]
```

```
corr_matrix = numeric_df.corr()
corr_matrix
```

	Age	Education	Usage	Fitness	Income	Miles	Price
Age	1.000000	0.280496	0.015064	0.061105	0.513414	0.036618	0.029263
Education	0.280496	1.000000	0.395155	0.410581	0.625827	0.307284	0.563463
Usage	0.015064	0.395155	1.000000	0.668606	0.519537	0.759130	0.623124
Fitness	0.061105	0.410581	0.668606	1.000000	0.535005	0.785702	0.696616
Income	0.513414	0.625827	0.519537	0.535005	1.000000	0.543473	0.695847
Miles	0.036618	0.307284	0.759130	0.785702	0.543473	1.000000	0.643923